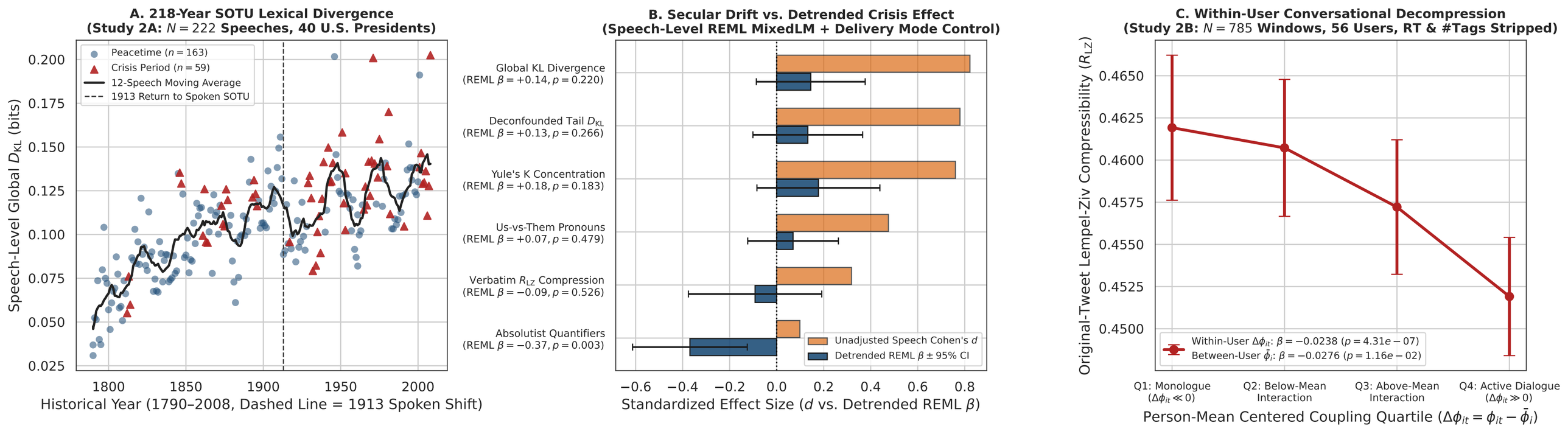

A. 218-Year SOTU Lexical Divergence
(Study 2A: N = 222 Speeches, 40 U.S. Presidents)
Peacetime (n = 163)
Crisis Period (n = 59)
12-Speech Moving Average
1913 Return to Spoken SOTU
Speech-Level Global D_{KL} (bits)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
1800
1850
1900
1950
2000
Historical Year (1790–2008, Dashed Line = 1913 Spoken Shift)
B. Secular Drift vs. Detrended Crisis Effect
(Speech-Level REML MixedLM + Delivery Mode Control)
Global KL Divergence
(REML $\beta = +0.14, p = 0.220$)
Deconfounded Tail D_{KL}
(REML $\beta = +0.13, p = 0.266$)
Yule's K Concentration
(REML $\beta = +0.18, p = 0.183$)
Us-vs-Them Pronouns
(REML $\beta = +0.07, p = 0.479$)
Verbatim R_{LZ} Compression
(REML $\beta = -0.09, p = 0.526$)
Absolutist Quantifiers
(REML $\beta = -0.37, p = 0.003$)
Unadjusted Speech Cohen's d
Detrended REML $\beta \pm 95\%$ CI
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
Standardized Effect Size (d vs. Detrended REML β)
C. Within-User Conversational Decompression
(Study 2B: N = 785 Windows, 56 Users, RT & #Tags Stripped)
Original-Tweet Lempel-Ziv Compressibility (R_{LZ})
0.4650
0.4625
0.4600
0.4575
0.4550
0.4525
0.4500
Within-User $\Delta\phi_{it}$: $\beta = -0.0238$ ($p = 4.31e-07$)
Between-User $\bar{\phi}_i$: $\beta = -0.0276$ ($p = 1.16e-02$)
Q1: Monologue
($\Delta\phi_{it} \ll 0$)
Q2: Below-Mean
Interaction
Q3: Above-Mean
Interaction
Q4: Active Dialogue
($\Delta\phi_{it} \gg 0$)
Person-Mean Centered Coupling Quartile ($\Delta\phi_{it} = \phi_{it} - \bar{\phi}_i$)